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Department of Computer Science

Master's in data science & business informatics

Distributed Data Analysis and Minig

Part 1

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Academic Year 2023/24

CONTEN

1 Introduction: 3

2 Data understanding and preparation: 3

2 Exploratory Data Analysis of Asteroid: 3

2.1.1 Asteroid Class Distribution: 3

2.1.2 Top 10 Most Destructive & Impact Severities Asteriods:……..…….4

2.1.3 Distribution of Hazardous Asteroids by Orbiting Body:……..4

2.1.4 Asteroid Behavioral Patterns: Orbital Clustering, Composition, and Anomalies……5

2.2 Data Cleaning: 6

2.2.1 Dimensionality Reduction: 6

2.2.2 Handling Missing Values: 7

3 Classification: 8

3.1 Features Importance on PHA : ……..... 8

3.2 Classification on Imbalanced Data: 8

3.2.1 Performance of Models on Imbalanced Data: 9

3.3 Classification on balanced Data: 10

3.3.1 Performance of Models on balanced Data: …….....10

4 Clustering: 12

4.1 K-means: 13

4.2 Gaussian Mixture Model (GMM): 14

4.3 Bisecting K-Means (BKM): 14

5. Conclusion: 16

1 Introduction:

This project focuses on the analysis of the NASA Jet Propulsion Laboratory's Asteroid Dataset, which includes detailed information on nearly a million asteroids. The dataset provides key insights into orbital parameters, physical properties, and the spatial positions of asteroids, offering a valuable resource for scientific research. The primary objectives are to preprocess the data by addressing missing values and eliminating irrelevant or redundant features, ensuring its readiness for analysis. Clustering techniques are applied to explore patterns and categorize asteroids into orbital classes, revealing potential groupings based on shared characteristics. classification models are developed to predict whether an asteroid poses a potential hazard to Earth, with a particular focus on overcoming challenges related to class imbalance. This comprehensive approach aims to enhance our understanding of asteroid characteristics and their potential risks.

2 Data understanding and preparation:

The dataset consisted of ***958,524 rows and 45 columns***, capturing detailed information about asteroid characteristics, including orbital parameters, physical properties, and hazard indicators. Preprocessing steps were taken to handle missing values, remove redundant features, and prepare the data for analysis. The goal was to ensure data consistency while retaining valuable information for clustering and classification tasks.

2.1 Exploratory Data Analysis of Asteroid

2.1.1 Asteroid Class Distribution

This pie chart illustrates the distribution of asteroid classes, highlighting that Main Belt Asteroids (MBA) dominate the dataset, constituting the majority. In contrast, other classes, such as the Outer Main Belt (OMB) and Inner Main Belt (IMB), represent a smaller portion. The significant dominance of MBA underscores their prevalence in the structure of our solar system.

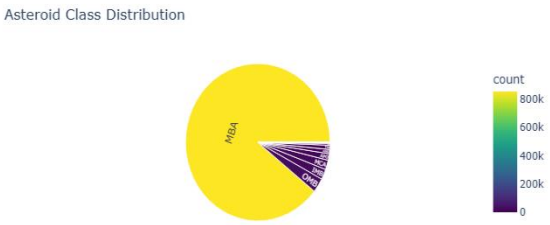
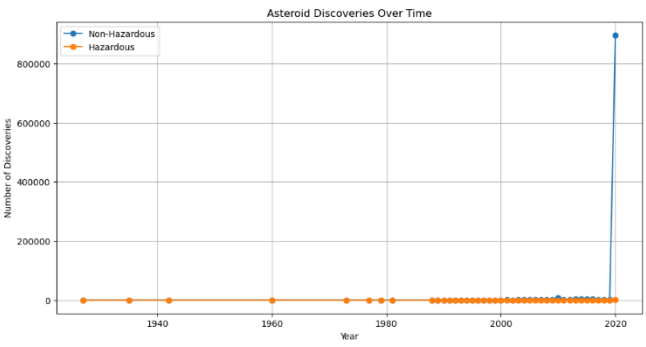


Figure: Asteroid Class Distribution



Next, we discover trends in asteroid discoveries over time, categorizing them into hazardous and non-hazardous groups. A noticeable surge in discoveries is evident in the late 20th century, correlating with advancements in astronomical technology and detection methods. While hazardous asteroid

Figure: Asteroid Discoveries Over Time

discoveries remain relatively low; their identification is crucial for planetary defense. The chart highlights the significant focus on cataloging non-hazardous asteroids, which dominate the dataset, reflecting the broader efforts in understanding our solar system's dynamics.

2.1.2 Top 10 Most Destructive & Impact Severities Asteroids:

In the Exploratory Data Analysis, an interesting discovery is revealed through the comparison of asteroid impact metrics. Below the figure **Top 10 Most Destructive Asteroids**, highlights asteroids with the highest calculated impact energy, with Varuna and Ceres standing out due to their significantly greater potential for devastation. The second figure, **"Top 10 Asteroid Impact Severities"**, complements this by focusing on impact severity, identifying Iris and Juno as particularly hazardous. By presenting these metrics side by side, the analysis provides a comprehensive perspective on asteroid threats, emphasizing the importance of monitoring and prioritizing potentially hazardous asteroids. This dual view ensures a holistic understanding of both destructive energy and severity, aiding in effective risk assessment.

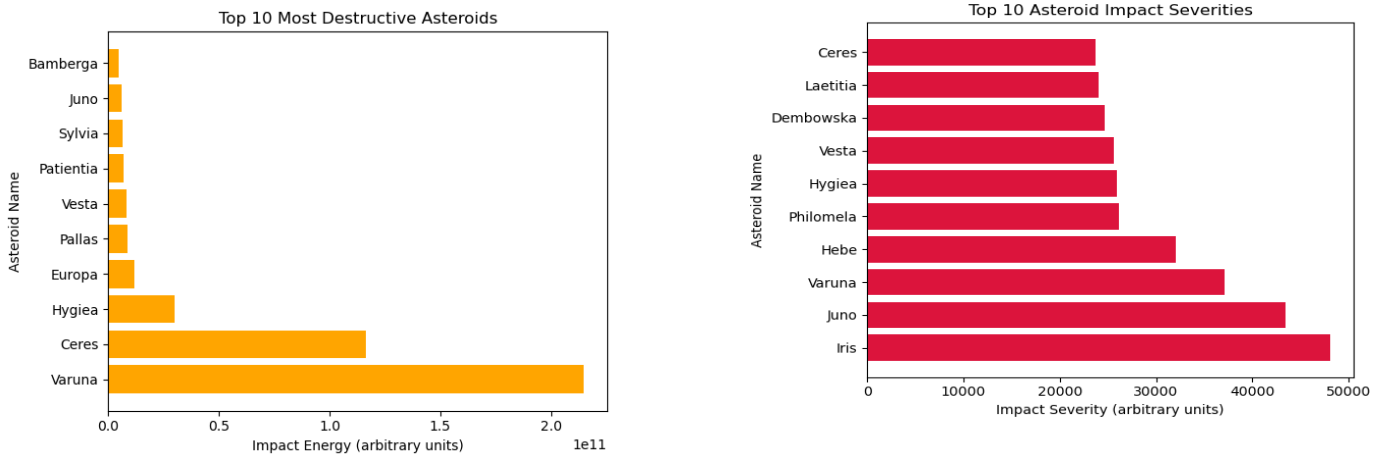
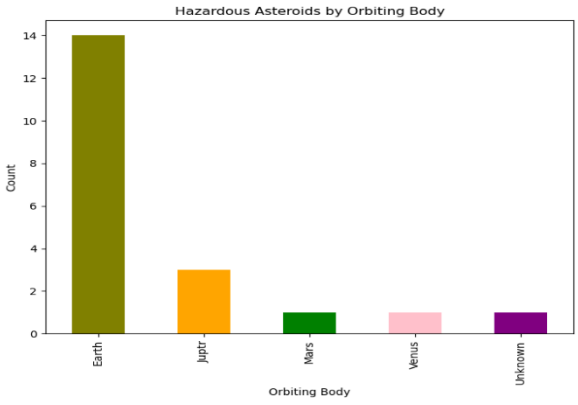


Figure: Top 10 Most Destructive Asteroids & Asteroid Impact Severities

2.1.3 Distribution of Hazardous Asteroids by Orbiting Body:

Hazardous asteroids exhibit varying orbiting patterns, as highlighted in the chart. Earth emerges as the primary orbital body with the highest count of hazardous asteroids, emphasizing the need for its constant monitoring. Jupiter follows with a smaller share, while Mars, Venus, and unknown orbiting bodies contribute marginally. This distribution underlines the critical focus required on Earth-



centric trajectories to address potential threats and ensure planetary safety.

Figure: Asteroids by Orbiting Body

Further, we observe the relationship between asteroid velocity and size across various classes, as showcased in the graph, which highlights unique behavioral patterns among them. The interactive plot emphasizes that certain asteroid classes, such as the Main Belt Asteroids (MBA), dominate the dataset with consistent velocity ranges, while larger and faster asteroids are rare but notable due to their potential impact risks.

This distribution underscores the importance of monitoring both the size and velocity of asteroids, as these factors significantly contribute to their potential hazard levels. The differentiation among asteroid classes also provides valuable insights into their formation and orbital dynamics, serving as a critical tool for further exploration and threat assessment.

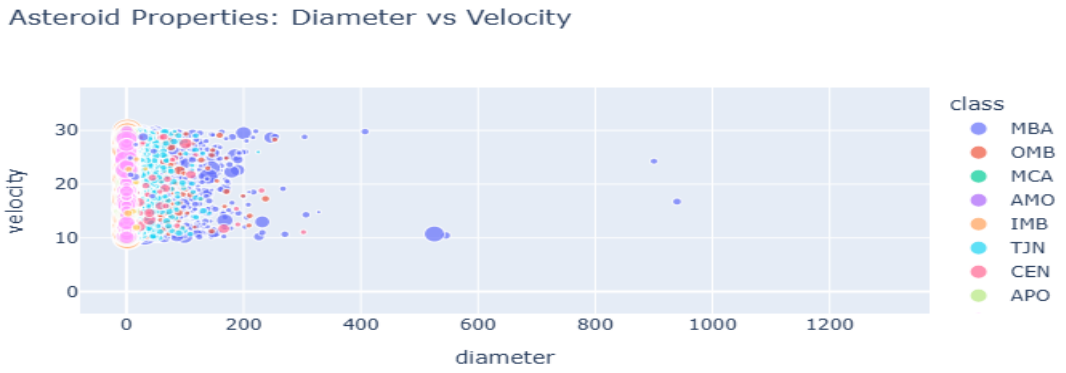


Figure: Asteroid Properties Diameter VS Velocity

2.1.4 Asteroid Behavioral Patterns: Orbital Clustering, Composition, and Anomalies:

The analysis integrates asteroid properties to uncover unique insights through clustering and anomaly detection. The 3D orbital clustering chart reveals how asteroids group based on their orbital parameters, such as semi-major axis, eccentricity, and inclination, highlighting distinct dynamical clusters. The composition-based clustering chart further emphasizes the role of albedo and diameter in distinguishing asteroid groups, offering valuable insights into their material makeup and potential origins. Additionally, the anomaly detection plot sheds light on outliers within the dataset, where exceptionally large or fast-moving asteroids

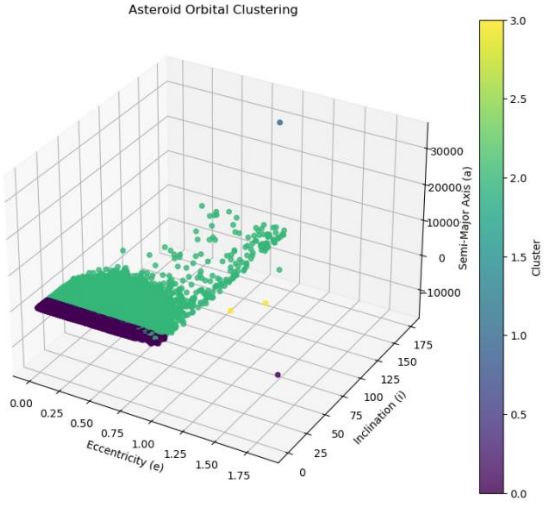


Figure: Asteroids by Orbital Clustering

deviate significantly from typical patterns, indicating their potential risk or unique formation history. Together, these visualizations provide a holistic view of asteroid behaviors, enabling a deeper understanding of their dynamics, composition, and outliers for more informed threat assessments and exploration strategies.

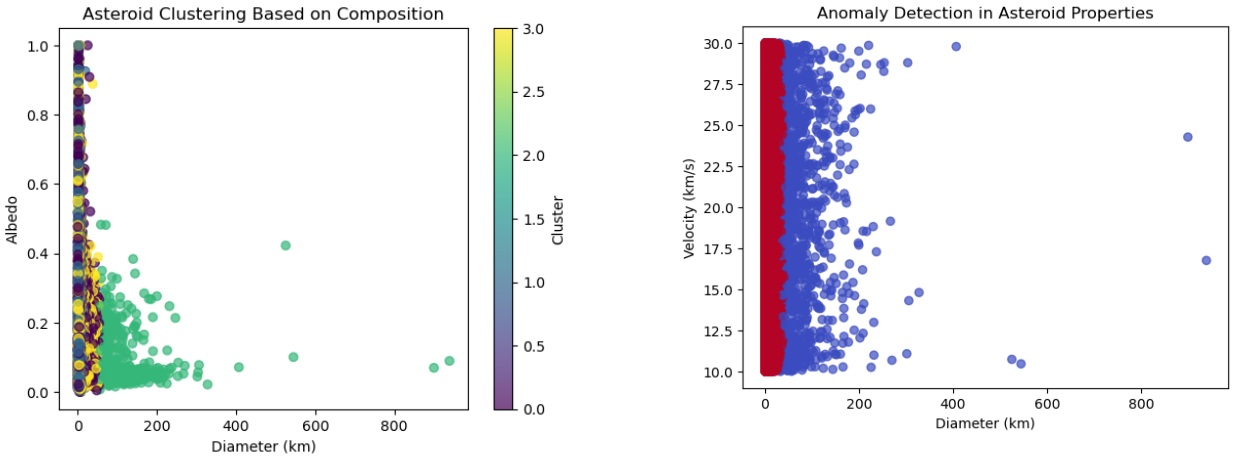


Figure: Asteroid clustering Composition, and Anomalies

During the analysis, a notable observation was the trend of asteroid close approaches over time. Although not included as a dedicated graph, the data revealed variations in close approaches across decades, with some years exhibiting significant peaks. This insight highlights the importance of tracking and monitoring asteroids for understanding their trajectories and potential threats.

2.2 Data Cleaning:

2.2.1 Dimensionality Reduction:

A Pearson correlation analysis was conducted to reduce redundancy and ensure the efficiency of the dataset. Variables with a correlation coefficient above 0.9 were considered for removal. However, no such variables were identified, confirming that all features contribute unique information. Moderate correlations were observed, such as between diameter_sigma and diameter (0.49), and between H (absolute magnitude) and n (mean motion, 0.46),



Figure I Correlation Matrix

but these were not strong enough to warrant exclusion. Weak correlations for features such as i (inclination) and epoch (orbital time) indicated their independence.

2.2.2 Handling Missing Values: The dataset was carefully examined to identify and address missing values, ensuring its readiness for analysis. For variables with negligible missing values, such as neo, ma, and rms, the affected rows were directly removed without significantly impacting the dataset. Similarly, records with missing values in the target variable pha (19,921 instances) were deleted. This step was essential because pha serves as the label for classification tasks, and rows without a valid target value could neither contribute to training nor be used for evaluation, rendering them unusable. For variables with substantial missing values, specific strategies were applied:

H (Absolute Magnitude): With 6,262 missing values, mean or median imputation was unsuitable due to the skewed nature of the distribution, as shown in Figure II (Distribution of H). Instead, the moderate correlation between H and n (mean motion) (R^2 = 0.34), as observed in Figure I (Correlation Matrix), was leveraged. A linear regression model was trained to predict the missing values of H, ensuring that the imputed values preserved the overall structure and relationships within the dataset.

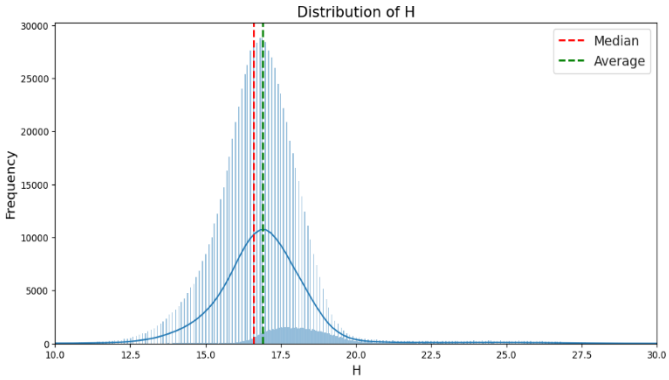


Figure II Distribution of H

Diameter: Missing values exceeded 85%, shows in Figure III (Distribution of Diameter) shows a relatively narrow distribution, making mean imputation appropriate as it closely represents the central tendency of the data.

Albedo: This variable exhibited a highly skewed distribution with no significant correlations to other features, making imputation unreliable. To avoid introducing bias or noise, Albedo was excluded entirely, ensuring that only relevant and robust data were retained.

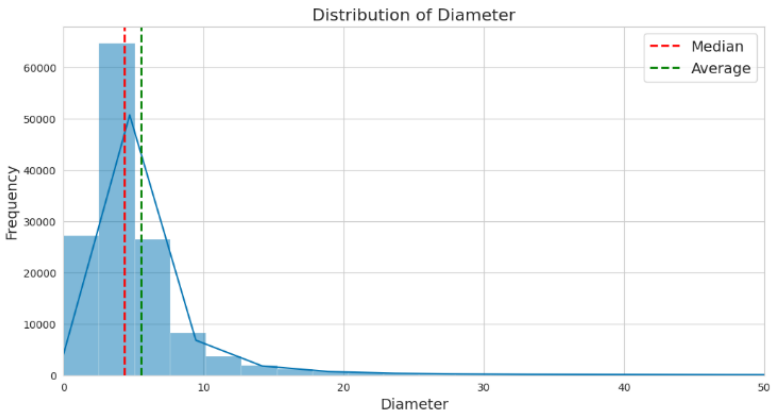


Figure III Distribution of Diameter

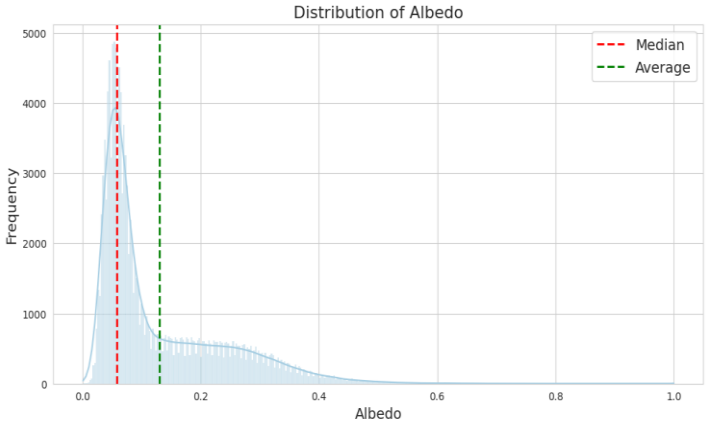


Figure IV Distribution of Albedo

After the data cleaning and preprocessing steps, including addressing missing values and removing irrelevant columns, the dataset was finalized with **938,597 records** and **17 columns**. These adjustments ensure that the data is complete, consistent, and well-prepared for further analysis.

3 Classification:

3.1 Feature Importance on Potential Hazardous Asteroid (PHA)

Before predicting, our goal of this analysis was to determine the significance of various features in predicting the Potentially Hazardous Asteroids (PHA). A Random Forest Classifier was employed to compute feature importance, which ranks the contribution of each feature to the prediction.

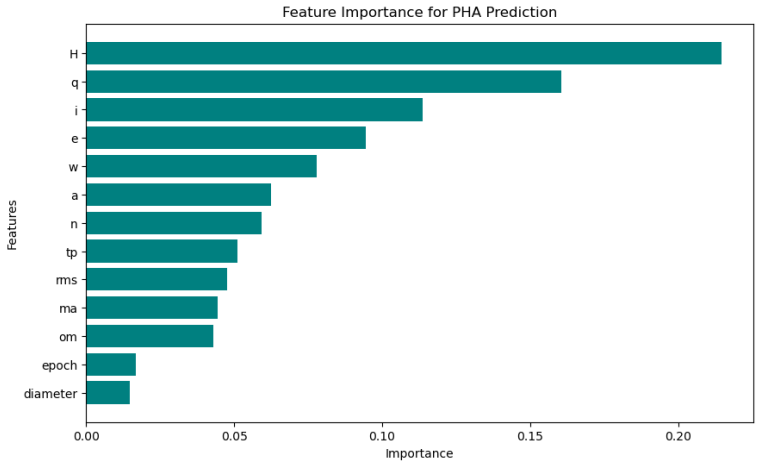


Figure: Feature Importance For PHA Prediction

Insights

- Top Contributors:
  - H (Absolute Magnitude): This feature had the highest importance, indicating its critical role in determining the PHA status.
  - q (Perihelion Distance) and i (Inclination): These orbital elements were also highly influential.
- Lesser Impact:
  - epoch (Observation Time) and diameter showed minimal contribution, suggesting limited influence on the model's decision-making process.

FEATURE	IMPORTANCE
H	0.214793
Q	0.160409
I	0.113800
E	0.094406
W	0.077822
A	0.062334
N	0.059146
TP	0.051205
RMS	0.047727
MA	0.044270
OM	0.042832
EPOCH	0.016623
DIAMETER	0.014632

Table: Feature Importance Values

3.2 Classification on Imbalanced Data:

The objective of this classification task was to develop a model capable of accurately predicting whether an asteroid poses a potential hazard to Earth (PHA). The dataset was divided into an 80% training set for model training and a 20% test set for evaluation. To assess model performance, both

the imbalanced dataset (original distribution) and a balanced dataset (using undersampling techniques) were used. Four classification models were implemented, Each model was chosen for its unique strengths, including handling imbalanced datasets, robustness, and capability with high-dimensional data.

During preprocessing low-predictive or categorical features such as rms, id, neo, and class were excluded, while the target variable pha (1 for hazardous and 0 for non-hazardous) was retained as the label for classification. The remaining features were vectorized using VectorAssembler and normalized with StandardScaler to ensure compatibility with the models. A significant class imbalance was identified, with the non-hazardous class (0) comprising 99.77% of the dataset.

The models were evaluated using key metrics, including Precision, Recall, F1 Score, AUC (Area Under the Curve), and Accuracy, alongside confusion matrices to provide detailed insights into predictions. This structured approach ensured robust evaluation of the models, highlighting their strengths and limitations when applied to both imbalanced and balanced scenarios. Detailed results and comparisons are presented in the subsequent sections.

3.2.1 Performance of Models on Imbalanced Data:

The models Random Forest Classifier (RFC), Logistic Regression (LR), Support Vector Machine (SVM), and Gradient Boosting (GBT) were trained and evaluated on the imbalanced dataset. Their performance metrics and confusion matrices are summarized below.

Model	Precision (Class 0)	Recall (Class 0)	Precision (Class 1)	Recall (Class 1)	AUC	Weighted F1	Accuracy
Random Forest	0.9977	1.0	0.0	0.0	0.9931	0.9988	0.9977
Logistic Regression	0.9981	0.9993	0.3073	0.1351	0.9966	0.9969	0.9975
Support Vector Machine	0.9978	1.0	0.0	0.0	0.9933	0.9967	0.9978
Gradient Boosting	0.9979	0.9999	0.8333	0.0253	0.9953	0.9969	0.9979

Confusion Matrices:

Random Forest Classifier (RFC):

	Predicted N	Predicted Y
Actual N	187,222	0
Actual Y	426	0

Logistic Regression (LR):

	Predicted N	Predicted Y
Actual N	**187,091**	**124**
Actual Y	**352**	**55**

Support Vector Machine (SVM):

	Predicted N	Predicted Y
Actual N	187,245	0
Actual Y	408	0

Gradient Boosting (GBT):

	Predicted N	Predicted Y
Actual N	187,261	2
Actual Y	386	10

The models demonstrated strong performance for the majority class (PHA = 0) but faced challenges predicting the minority class (PHA = 1) due to the extreme class imbalance:

- **Random Forest (RFC):** Achieved perfect recall for PHA = 0 but failed entirely for PHA = 1, with an AUC of 0.9931.
- **Logistic Regression (LR):** Showed some improvement for PHA = 1, with a precision of 30.73% and a recall of 13.51%, achieving an AUC of 0.9966.
- **Support Vector Machine (SVM):** Performed well for PHA = 0 but failed to predict any instances of PHA = 1, resulting in an AUC of 0.9933.
- **Gradient Boosting (GBT):** Achieved the highest precision for PHA = 1 (83.33%) but with a low recall of 2.53%. Its overall AUC was 0.9953.

These results highlight the models' strong performance for the majority class but significant struggles with the minority class, emphasizing the importance of balancing techniques to improve predictions for hazardous asteroids (PHA = 1).

3.3 Classification on balanced Data:

To address the extreme class imbalance in the dataset, a random undersampling technique was applied to balance the training data. This involved reducing the majority class (PHA = 0) to match the size of the minority class (PHA = 1), ensuring equal representation of both classes. This adjustment allowed models to focus equally on predicting hazardous asteroids (PHA = 1) and non-hazardous asteroids (PHA = 0), leading to more balanced performance.

Balancing significantly improved the recall for hazardous asteroids across all models, enabling them to identify more instances of PHA = 1. However, this improvement came at the cost of precision, as some models misclassified non-hazardous asteroids as hazardous, increasing false positives.

3.3.1 Performance of Models on Undersampled Data:

The models Random Forest Classifier (RFC), Logistic Regression (LR), Support Vector Machine (SVM), and Gradient Boosting (GBT) were retrained on the balanced dataset and evaluated on the original test set. The performance metrics for each model are summarized in the table below, along with the confusion matrices.

Model	Precision (Class 0)	Recall (Class 0)	Precision (Class 1)	Recall (Class 1)	AUC	Weighted Precision	Weighted Recall	Weighted F1
Random Forest	1.0000	0.9804	0.0987	1.0000	0.9980	0.9981	0.9805	0.9884
Logistic Regression	1.0000	0.9859	0.1403	1.0000	0.9969	0.9980	0.9859	0.9912
SVM	1.0000	0.9797	0.0960	1.0000	0.9966	0.9981	0.9798	0.9880
Gradient Boosting	0.9999	0.9843	0.1199	0.9975	0.9973	0.9981	0.9843	0.9904

Confusion Matrices:

Random Forest (RFC):

	Predicted N	Predicted Y
Actual N	183,601	3,663
Actual Y	0	401

Logistic Regression (LR):

	Predicted N	Predicted Y
Actual N	184,592	2,640
Actual Y	0	431

Support Vector Machine (SVM):

	Predicted N	Predicted Y
Actual N	183,430	3,795
Actual Y	0	403

Gradient Boosting (GBT):

	Predicted N	Predicted Y
Actual N	184,289	2,937
Actual Y	1	400

The retrained models showed significant improvement in identifying hazardous asteroids (PHA = 1) due to the balanced dataset, achieving high recall for the minority class.

- Random Forest (RFC): Achieved 100% recall for PHA = 1, but with a low precision of 9.87% due to false positives, resulting in a strong overall AUC of 0.9980.
- Logistic Regression (LR): Provided the highest precision for PHA = 1 (14.03%) and strong recall for PHA = 0 (98.59%), with an AUC of 0.9969.
- Support Vector Machine (SVM): Struggled with precision for PHA = 1 (9.60%) but maintained solid recall for PHA = 0 (97.97%) and an AUC of 0.9966.

- Gradient Boosting (GBT): Balanced precision (11.99%) and recall (99.75%) for PHA = 1 while maintaining excellent performance for the majority class, achieving an AUC of 0.9973.

The results highlight the trade-offs introduced by balancing the data. Random Forest and Support Vector Machine achieved 100% recall for hazardous asteroids but suffered from low precision due to false positives. In contrast, Gradient Boosting provided the best overall performance, striking a balance between precision and recall for PHA = 1 while maintaining strong metrics for the majority class.

These findings demonstrate the effectiveness of balancing techniques in improving minority class predictions and underscore the importance of evaluating both precision and recall addressing the challenges posed by class imbalance effectively.

4 Clustering:

The analysis investigated the distribution of orbital classes in the dataset, which comprises 12 unique categories corresponding to various asteroid orbital types. As illustrated in the graph showing the frequency of distinct values in the class column (logarithmic scale), MBA (Main Belt Asteroids) is the most dominant class, followed by OMB (Outer Main Belt), IMB (Inner Main Belt), and MCA (Mars-Crossing Asteroids). Less frequent classes include IEO (Interior Earth Objects) and AST (Asteroid).

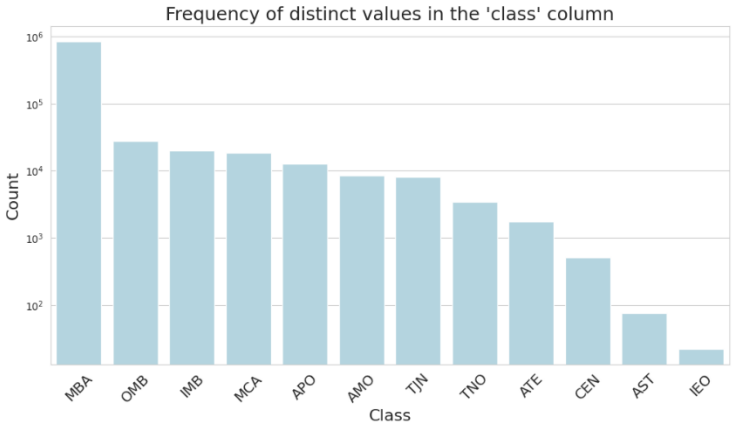


Figure V

Before applying the K-means clustering algorithm, preprocessing steps were conducted to prepare the data. Features such as rms, id, pha, neo, and class were excluded due to their categorical nature or lack of predictive value. The remaining features were vectorized and normalized using a Standard Scaler, ensuring compatibility with the clustering algorithm.

4.1 K-means

The K-means algorithm was implemented with data normalized using a Standard Scaler, which proved more effective than Min-Max scaling. The choice of K (number of clusters) was determined through an iterative approach, evaluating metrics such as SSE (Sum of Squared Errors) and the Silhouette Score for values of K ranging from 2 to 29.

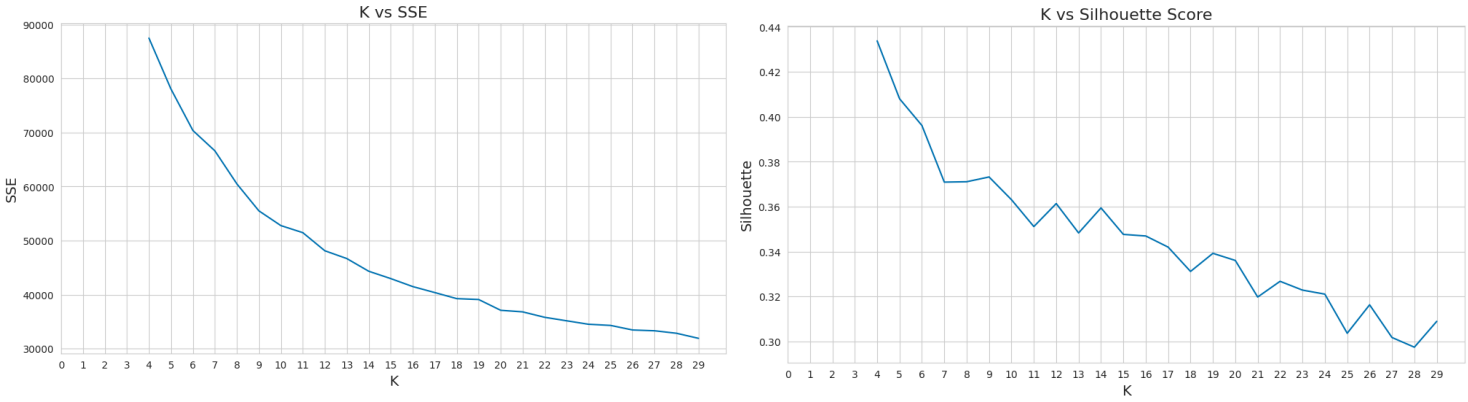


Figure VI

The SSE decreases as K increases, which is expected because additional clusters reduce the average distance between points and their centers.

The rate of SSE reduction significantly slows after K=6, forming a noticeable **"elbow point"**. This suggests that increasing K beyond this point yields diminishing returns in clustering improvement.

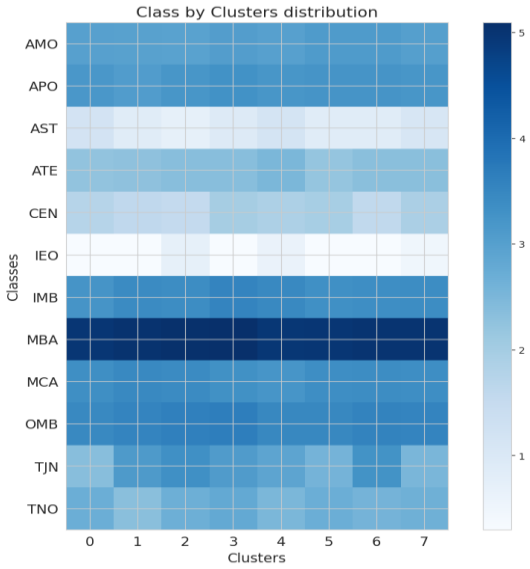
The Silhouette Score is highest at K=4 and decreases gradually as K increases.

For K>6, the score continues to drop, reflecting declining clustering quality as clusters become less distinct.

At K=6, the Silhouette Score is lower than at K=4, but still acceptable, indicating reasonably well-separated clusters.

Based on both graphs, K=4 is optimal for maximizing cluster separation (Silhouette Score) and k=6 balance SSE reduction (compact clusters) and moderate separation. K=6 appears to be a more practical choice based on SSE and Silhouette score trends.

The heatmap analysis illustrates the distribution of predefined orbital classes within the identified clusters. The clustering results revealed that the clusters do not align strongly with predefined orbital classes, suggesting that K-means captured other latent patterns in the dataset. For example, while some clusters contained higher concentrations of specific classes, such as TNO, these classes were still distributed across multiple clusters.



This analysis highlights that the clustering process likely represents underlying patterns or features in the dataset that extend beyond predefined orbital classifications. These findings underscore the potential for further exploratory analysis to uncover additional relationships among asteroids, providing deeper insights into their groupings and characteristics.

4.2 Gaussian Mixture Model (GMM)

We used the Gaussian Mixture Model (GMM) to group (or cluster) asteroids based on their features. GMM clustering organizes asteroids into meaningful groups, revealing trends and patterns in their features. The heatmap makes it easy to spot which classes are more common in certain clusters and provides a foundation for deeper exploration into asteroid characteristics. We used this to handle overlapping clusters better than K-means.

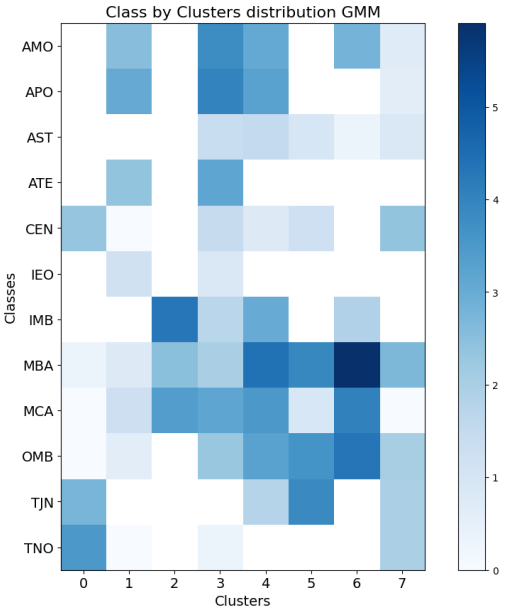


Figure VIII

In heatmap, Each row represents a specific asteroid class (e.g., Main-Belt Asteroids (MBA) or Mars Crossing Asteroids (MCA)). Each column represents one of the 8 clusters identified by the GMM. The darker the cell, the more asteroids of that class belong to that cluster (after applying a logarithmic scale to the counts to make the data easier to see).

4.3 Bisecting K-Mean (BKM)

BKM was applied to scaled asteroid data with 8 clusters. The algorithm splits data iteratively to ensure a hierarchical division while reducing the intra-cluster variation.

The heatmap represents the log-transformed counts of asteroid classes across clusters. A logarithmic transformation (log10) is used to manage wide-ranging values and improve interpretability.

According to heatmap, the dominance of certain classes like MBA in multiple clusters highlights the variability within this class and suggests the need for further sub-grouping within MBA. The distinct presence of classes like IMB and MCA in specific clusters shows the model's ability to separate data points based on subtle differences in features.

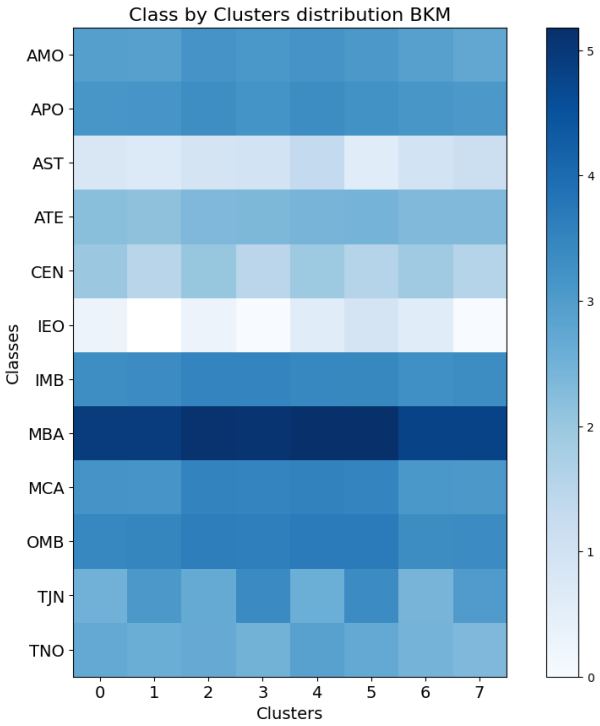


Figure IX

Cluster Predictions Comparison Table of Top 20 Rows

Index	GMM Prediction	BKM Prediction	K-Means Prediction
1	7	2	1
2	5	7	4
3	5	7	4
4	7	6	7
5	5	6	7
6	5	1	3
7	5	4	0
8	5	2	1
9	5	1	3
10	5	4	0
11	5	5	2
12	6	5	3
13	5	1	1
14	6	3	3
15	6	1	2
16	7	5	5
17	5	2	0
18	6	4	3
19	5	1	3
20	5	0	3

Tables II

5. Conclusion

This report successfully analysed a comprehensive asteroid dataset to assess whether an asteroid poses a potential hazard to Earth (PHA). Through effective preprocessing, missing values were addressed, irrelevant features were removed, and the dataset's integrity was maintained for analysis. Classification models demonstrated strong performance for predicting non-hazardous asteroids (PHA = 0) but faced challenges in identifying hazardous ones (PHA = 1) due to class imbalance. By employing balancing techniques, predictions for hazardous asteroids improved significantly, with Gradient Boosting achieving the best trade-off between precision and recall.

Clustering analysis revealed distinct groupings among asteroids but showed no significant alignment with predefined orbital classes, suggesting that the clusters represent other latent patterns within the dataset. These findings demonstrate the effectiveness of preprocessing, classification, and clustering techniques in analyzing asteroid data and enhancing understanding of their characteristics and potential risks.